

Survival MF Derivations

Andrew Walther

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1 Introduction

The following document presents derivations for matrix factorization & survival expressions for supervised matrix factorization. Content includes an expression of the overall objective function to minimize as well as the procedure for updating the prior distributions for L, F, β . These updates eventually are solved via the EBNM problem illustrated in the EBMF paper.

1A Models

The following models are considered as part of this supervised matrix factorization task:

- Genomics Data: $Y_{n \times p} = L_{n \times k} F'_{k \times p} + E_{n \times p}$, where $E_{ij} \stackrel{\text{iid}}{\sim} N(0, \tau_j^{-1})$
- Survival Data: $h(t_i) = h_0(t_i) \exp(\sum_k l_{ik} \beta_k)$

1B Update Procedure

The general procedure for parameter (prior & posterior distribution) updates is as follows.

For $b=1, \dots, B$

1. Iterate over $k = 1, \dots, K$ for each of the following.
2. update $q(l_k), g(l_k)$ with other parameters fixed (different from EBMF problem)
3. update $q(f_k), g(f_k)$ with other parameters fixed (identical to EBMF problem)
4. update $q(\beta_k), g(\beta_k)$ with other parameters fixed (different from EBMF problem)
5. update τ_{ij} (identical to EBMF problem)

2 Objective Function

The objective function to minimize that considers priors (g) & posteriors (q) for all parameters is

$$F \left(\underbrace{q_l, q_f, q_\beta}_{\text{posteriors}}, \underbrace{g_l, g_f, g_\beta}_{\text{priors}}, \tau \right) = E_{q_l, q_f, q_\beta} \left[\log P \left(\underbrace{Y, t, \delta}_{\text{observed}} \mid \underbrace{L, F, \beta, \tau}_{\text{unknown}} \right) \right] \\ + \sum_k E_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] + \sum_k E_{q_{f_k}} \left[\log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right] + \sum_k E_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \quad (1)$$

and the full data likelihood is expressed as

$$\log P \left(\underbrace{Y}_{\text{Genomics}}, \underbrace{t, \delta}_{\text{Survival}} \mid L, F, \beta, \tau \right) = \underbrace{\log P(Y \mid L, F, \tau^{-1})}_{\text{Genomics}} + \underbrace{\log P(t, \delta \mid L, \beta)}_{\text{Survival(new)}} \quad (2)$$

3 Taylor Series Approximation of Survival Likelihood

Now we consider a 2nd-Order Taylor Series Expansion of the survival data likelihood for the sake of finding a tractable solution.

First, note that the 2nd-Order approximation takes the form

$$f(x) = f(x_0) + \underbrace{f'(x_0)}_{\text{gradient/score}} (x - x_0) + \frac{1}{2} \underbrace{f''(x_0)}_{\text{hessian/info}} (x - x_0)^2 \quad (3)$$

and that the linear predictor for the survival data has the form

$$\eta_i = l'_i \beta = \sum_k l_{ik} \beta_k \quad (4)$$

Then the Approximation is as follows

assume non-informative prior -> get Cox model

$$\log P(t, \delta | L, \beta) = l(\eta) \quad (5)$$

$$\approx l(\hat{\eta}) + (\eta - \hat{\eta})^\top \underbrace{\nabla l(\hat{\eta})}_{\mathbf{U}} + \frac{1}{2} (\eta - \hat{\eta})^\top \underbrace{\nabla^2 l(\hat{\eta})}_{\mathbf{H} = -\mathbf{W}} (\eta - \hat{\eta}) \quad (6)$$

$$\approx l(\hat{\eta}) + \sum_i (\eta_i - \hat{\eta}_i) \mathbf{u}_i + \frac{1}{2} \sum_i (\eta_i - \hat{\eta}_i)^2 \mathbf{H}_{ii} \quad (\text{quadratic form: } \sum_i ax_i^2 + bx_i + c \rightarrow a=?, b=?) \quad (7)$$

$$= \sum_i \left[\mathbf{u}_i (\eta_i - \hat{\eta}_i) - \frac{1}{2} \mathbf{W}_{ii} (\eta_i^2 - 2\eta_i \hat{\eta}_i + \hat{\eta}_i^2) \right] \quad (8)$$

$$= \sum_i \left[\mathbf{u}_i \eta_i - \mathbf{u}_i \hat{\eta}_i - \frac{1}{2} \mathbf{W}_{ii} \eta_i^2 + \mathbf{W}_{ii} \eta_i \hat{\eta}_i - \frac{1}{2} \mathbf{W}_{ii} \hat{\eta}_i^2 \right] \quad (\text{keep only terms with } \eta_i) \quad (9)$$

$$\text{where } \sum_i (a_i \eta_i^2 + b_i \eta_i) = \sum_i a_i \left(\eta_i + \frac{1}{2} \frac{b_i}{a_i} \right)^2 \quad (10)$$

$$= \sum_i \left[\left(\underbrace{-\frac{1}{2} \mathbf{W}_{ii}}_A \right) \eta_i^2 + \left(\underbrace{\mathbf{u}_i + \mathbf{W}_{ii} \hat{\eta}_i}_B \right) \eta_i \right] \quad (11)$$

$$= \sum_i \left(-\frac{1}{2} \mathbf{W}_{ii} (\eta_i - c_i)^2 \right) \quad (12)$$

$$\text{where } B = \mathbf{W}_{ii} c_i = \mathbf{u}_i + \mathbf{W}_{ii} \hat{\eta}_i \Rightarrow c_i = \frac{\mathbf{u}_i}{\mathbf{W}_{ii}} + \hat{\eta}_i \quad (13)$$

$$= \sum_i \left(-\frac{1}{2} \mathbf{W}_{ii} \eta_i^2 + \mathbf{W}_{ii} \eta_i c_i - \frac{1}{2} \mathbf{W}_{ii} c_i^2 \right) \quad (14)$$

$$= \sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(\eta_i - \left(\underbrace{\frac{\mathbf{u}_i}{\mathbf{W}_{ii}} + \hat{\eta}_i}_c \right) \right)^2 \right] \quad (15)$$

4 Update for q_l

For the updates of q_l & g_l , we consider terms from F that pertain to l . Therefore, we consider

$$F(q_l, g_l) = E_{q_l, q_f, q_\beta} [\log P(Y, t, \delta | L, F, \beta, \tau)] + \sum_k E_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] \quad (16)$$

$$= \underbrace{E_{q_l, q_f, q_\beta} [\log P(Y | L, F, \tau^{-1})]}_{\text{Equation 1}} + \underbrace{E_{q_l, q_f, q_\beta} [\log P(t, \delta | L, \beta)]}_{\text{Equation 2}} + \underbrace{\sum_k E_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right]}_{\text{Keep from above}} \quad (17)$$

$$= \sum_k \left[\text{Terms from 1}(l_k) + \text{Terms from 2}(l_k) + E_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] \right] \quad (18)$$

Now, we need expressions for the terms of equations 1 & 2 that are dependent on l_k and/or l_k^2 . To aid this process, we first consider the residuals for y_{ij} , denoted as R_{ij} :

$$R_{ij} = y_{ij} - \sum_k l_{ik} f_{jk} \quad (19)$$

$$= y_{ij} - \left[l_{ik} f_{jk} + \sum_{k \neq k'} l_{ik'} f_{jk'} \right] \quad (20)$$

To consider the k -th residual, also consider rearranging this result such that

$$R_{ij} = \left(\underbrace{y_{ij} - \sum_{k \neq k'} l_{ik'} f_{jk'}}_{R_{ij}^k} \right) - l_{ik} f_{jk} \quad (21)$$

where the k -th residual is $R_{ij}^k = y_{ij} - \sum_{k' \neq k} l_{ik'} f_{jk'}$. Similarly, residuals for the k -th residual of z_i are expressed as

$$z_i^{-k} = z_i - \sum_{k' \neq k} l_{ik'} \beta_{k'}. \quad (22)$$

Note: the EBMF paper uses R_{ij}^k to represent the posterior first moment ($\overline{l_{ik'}}$ and $\overline{f_{jk'}}$) of the residual, but we use it to represent the residual itself.

For example, we have

$$E_{q_l, q_f} [R_{ij}^k] = y_{ij} - \sum_{k \neq k'} \overline{l_{ik'} f_{jk'}} \quad (23)$$

$$= E_{q_l, q_f} \left[y_{ij} - \sum_{k \neq k'} l_{ik'} f_{jk'} \right] \quad (24)$$

$$= E_{q_l, q_f} [y_{ij}] - \sum_{k \neq k'} E_{q_l, q_f} [l_{ik'} f_{jk'}] \quad (25)$$

$$= y_{ij} - \sum_{k \neq k'} E_{q_l, q_f} [l_{ik'}] E_{q_l, q_f} [f_{jk'}] \quad (26)$$

$$= y_{ij} - \sum_{k \neq k'} \overline{l_{ik'} f_{jk'}} \quad (27)$$

We can now use these results to simplify the procedure of identifying the necessary terms of equations 1 & 2 to specify coefficients A and B that are attached to the terms l_{ik} and l_{ik}^2 .

First consider the log-likelihood term

$$E_{q_l, q_f, q_\beta} [\log P(Y|L, F, \tau^{-1})] = E_{q_l, q_f, q_\beta} \left[\log \left(\prod_{ij} \frac{\tau_{ij}}{\sqrt{2\pi}} \exp \left[-\frac{\tau_{ij}}{2} \left(y_{ij} - \underbrace{\sum_k l_{ik} f_{jk}}_Y \right)^2 \right] \right) \right] \quad (28)$$

where the observed values are $y_{ij} = \sum_k l_{ik} f_{jk} + \varepsilon_{ij}$ and $\varepsilon \sim N(0, \sigma^2 = \tau^{-1})$

Equation 1 is then expressed as

$$E_{q_l, q_f, q_\beta} [\log P(Y|L, F, \tau^{-1})] = E_{q_l, q_f, q_\beta} \left[\sum_{ij} \left[-\frac{1}{2} \tau_{ij} \left(y_{ij} - \sum_k l_{ik} f_{jk} \right)^2 \right] \right] \quad (29)$$

$$= E_{q_l, q_f, q_\beta} \left[\sum_{ij} \left[-\frac{1}{2} \tau_{ij} (R_{ij}^k - l_{ik} f_{jk})^2 \right] \right] \quad (30)$$

$$= E_{q_l, q_f, q_\beta} \left[\sum_{ij} \left[-\frac{1}{2} \tau_{ij} \left(R_{ij}^{k^2} - 2R_{ij}^k l_{ik} f_{jk} + l_{ik}^2 f_{jk}^2 \right) \right] \right] \quad (31)$$

Similarly, equation 2 is expressed as

$$E_{q_l, q_f, q_\beta} [\log P(t, \delta|L, \beta)] = E_{q_l, q_f, q_\beta} \left[\sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(\underbrace{\sum_k l_{ik} \beta_k}_{\eta_i} - \underbrace{\left(\frac{\mathbf{u}_i}{\mathbf{W}_{ii}} + \hat{\eta}_i \right)}_{z_i} \right)^2 \right] \right] \quad (32)$$

$$= E_{q_l, q_f, q_\beta} \left[\sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(z_i - \sum_k l_{ik} \beta_k \right)^2 \right] \right] \quad (33)$$

$$= E_{q_l, q_f, q_\beta} \left[\sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} (z_i^k - l_{ik} \beta_k)^2 \right] \right] \quad (34)$$

$$= E_{q_l, q_f, q_\beta} \left[\sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(z_i^{k^2} - 2z_i^k l_{ik} \beta_k + l_{ik}^2 \beta_k^2 \right) \right] \right] \quad (35)$$

Now, we want to sum these two results and identify the coefficients A and B for the terms l_{ik} and l_{ik}^2 that satisfy the expression $A \cdot l_{ik}^2 + B \cdot l_{ik}$. It follows that (considering the result of equation 18)

$$\begin{aligned} F(q_l, g_l) &= \sum_k \left[E_{q_l, q_f, q_\beta} \left[\sum_{ij} \left[-\frac{1}{2} \tau_{ij} \left(R_{ij}^{k^2} - 2R_{ij}^k l_{ik} f_{jk} + l_{ik}^2 f_{jk}^2 \right) \right] \right] \right. \\ &\quad \left. + \sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(z_i^{k^2} - 2z_i^k l_{ik} \beta_k + l_{ik}^2 \beta_k^2 \right) \right] \right] \\ &\quad + E_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] \end{aligned} \quad (36)$$

Then for the genomics data (equation 1), the coefficients are

- l_{ik} coefficient (B): $\tau_{ij} R_{ij}^k E[f_{jk}] = \tau_{ij} R_{ij}^k \overline{f_{jk}}$
- l_{ik}^2 coefficient (A): $\tau_{ij} E[f_{jk}^2] = \tau_{ij} \overline{f_{jk}^2}$

and for the survival data (equation 2), the coefficients are

- l_{ik} coefficient (B): $\mathbf{W}_{ii} z_i^k E[\beta_k] = \mathbf{W}_{ii} z_i^k \overline{\beta_k}$
- l_{ik}^2 coefficient (A): $\mathbf{W}_{ii} E[\beta_k^2] = \mathbf{W}_{ii} \overline{\beta_k^2}$

such that the combined coefficients are

$$\mathbf{A}_{ik} = \sum_j \tau_{ij} \overline{f_{jk}^2} + \mathbf{W}_{ii} \overline{\beta_k^2} \quad (37)$$

$$\mathbf{B}_{ik} = \sum_j \tau_{ij} R_{ij}^k \overline{f_{jk}} + \mathbf{W}_{ii} z_i^k \overline{\beta_k} \quad (38)$$

Then, for the EBNM problem with $EBNM(x, s)$ where $x_i = \frac{B_{ik}}{A_{ik}}$ and $s_i^2 = \frac{1}{A_{ik}}$, we have that

$$x_i = \frac{\sum_j \tau_{ij} R_{ij}^k \overline{f_{jk}} + \mathbf{W}_{ii} z_i^k \overline{\beta_k}}{\sum_j \tau_{ij} \overline{f_{jk}^2} + \mathbf{W}_{ii} \overline{\beta_k^2}} \quad (39)$$

$$s_i^2 = \frac{1}{\sum_j \tau_{ij} \overline{f_{jk}^2} + \mathbf{W}_{ii} \overline{\beta_k^2}} \quad (40)$$

and the EBNM problem can be solved via the EBNM R package.

5 Update for q_f

Updates for the prior & posterior of F follow the process as expressed in the EBMF paper. We want to find an update for the variational posterior $q_{f_k}(f_k)$ and prior g_{f_k} that maximize the objective function $F(\cdot)$ with all other parameters held fixed. To write out the update for q_f , note that F is only present in the term in the objective function for the genomics data (and isn't in the survival data term) such that we consider

$$F(q_{f_k}) \approx E_{q_l, q_f, q_\beta} [\log P(Y|L, F, \tau)] + E_{q_l, q_f, q_\beta} [\log g_{f_k}(f_k) - \log q_{f_k}(f_k)]. \quad (41)$$

The genomics data model is

$$Y_{ij} \sim N\left(\sum_k l_{ik} f_{jk}, \tau_{ij}^{-1}\right). \quad (42)$$

Then expand the log-likelihood to be

$$E_{q_l, q_f, q_\beta} [\log P(Y|L, F, \tau)] = \sum_{ij} E_{q_l, q_f, q_\beta} \left[-\frac{1}{2} \tau_{ij} \left(y_{ij} - \sum_k l_{ik} f_{jk} \right)^2 \right] \quad (43)$$

and isolate the k th factor by defining the residual excluding factor k as

$$R_{ij}^k = y_{ij} - \sum_{k \neq k'} l_{ik'} f_{jk'} \quad (44)$$

such that

$$R_{ij} = \left(\underbrace{y_{ij} - \sum_{k \neq k'} l_{ik'} f_{jk'}}_{R_{ij}^k} \right) - l_{ik} f_{jk}. \quad (45)$$

Then substitute this result back into the likelihood expression to get

$$E_{q_l, q_f, q_\beta} [\log P(Y|L, F, \tau)] = \sum_{ij} E_{q_l, q_f, q_\beta} \left[-\frac{1}{2} \tau_{ij} (R_{ij}^k - l_{ik} f_{jk})^2 \right]. \quad (46)$$

We can then expand the squared term to isolate f_{jk} in the likelihood

$$E_{q_l, q_f, q_\beta} [\log P(Y|L, F, \tau)] = \sum_{ij} E_{q_l, q_f, q_\beta} \left[-\frac{1}{2} \tau_{ij} \left((R_{ij}^k)^2 - 2R_{ij}^k l_{ik} f_{jk} + l_{ik}^2 f_{jk}^2 \right) \right]. \quad (47)$$

and consider the expectation $E_q[\cdot]$ with respect to L , treating the residual R_{ij}^k as fixed, so that

$$E_{q_l, q_f, q_\beta} [\log P(Y|L, F, \tau)] = \sum_{ij} E_{q_l, q_f, q_\beta} \left[-\frac{1}{2} \tau_{ij} \left[\cancel{(R_{ij}^k)^2} - 2R_{ij}^k l_{ik} f_{jk} + l_{ik}^2 f_{jk}^2 \right] \right]. \quad (48)$$

Now, group terms to identify coefficients for f and f^2 to get the quadratic form $-\frac{1}{2}Af^2 + Bf$ (summing over i for each j to update f_k)

$$F(q_{f_k}) \approx E_{q_l, q_f, q_\beta} \left[\sum_j \left[-\frac{1}{2} f_{jk}^2 \left(\underbrace{\sum_i \tau_{ij} l_{ik}^2}_{A_{jk}} \right) + f_{jk} \left(\underbrace{\sum_i \tau_{ij} R_{ij}^k l_{ik}}_{B_{jk}} \right) \right] \right] + E_{q_{f_k}} \left[\log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right]. \quad (49)$$

Therefore, we have

$$\mathbf{A}_{\mathbf{jk}} = \sum_i \tau_{ij} \overline{l_{ik}^2} \quad (50)$$

$$\mathbf{B}_{\mathbf{jk}} = \sum_i \tau_{ij} R_{ij}^k \overline{l_{ik}} \quad (51)$$

which yield the following inputs for the EBNM problem

$$x_i = \frac{\mathbf{B}_{\mathbf{jk}}}{\mathbf{A}_{\mathbf{jk}}} = \frac{\sum_i \tau_{ij} R_{ij}^k \overline{l_{ik}}}{\sum_i \tau_{ij} \overline{l_{ik}^2}} \quad (52)$$

$$s_i^2 = \frac{1}{\mathbf{A}_{\mathbf{jk}}} = \frac{1}{\sum_i \tau_{ij} \overline{l_{ik}^2}} \quad (53)$$

6 Update for q_β

First, note that for the updates of q_β , β only appears in the likelihood for the survival data, so we can focus on that term and ignore the likelihood term for the genomics data. The full objective function is

$$F(q_l, q_f, q_\beta, g_l, g_f, g_\beta, \tau) = E_{q_l, q_f, q_\beta} [\log P(Y, t, \delta|L, F, \beta, \tau)] + \sum_k E_{q_{l_k}} \left[\frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] \\ + \sum_k E_{q_{f_k}} \left[\frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right] + \sum_k E_{q_\beta} \left[\frac{g_\beta(\beta_k)}{q_\beta(\beta_k)} \right] \quad (54)$$

The full data likelihood is

$$\log P(Y, t, \delta|L, F, \beta, \tau) = \underbrace{\log P(Y|L, F, \beta, \tau^{-1})}_{\text{genomics}} + \underbrace{\log P(t, \delta|L, \beta)}_{\text{survival}} \quad (55)$$

such that the objective function for q_β is

$$F(q_\beta, g_\beta) = E_{q_l, q_f, q_\beta} [\log P(t, \delta|L, \beta)] + \sum_k E_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \quad (56)$$

For the likelihood with q_β , consider a 2nd order Taylor series approximation (similar to the update for q_l) where we have

$$\log P(t, \delta | L, \beta) = \ell(\eta) \quad (\text{where } \eta = l\beta) \quad (57)$$

$$\approx \dots (\text{derived in } q_l \text{ update}) \quad (58)$$

$$= \sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(\underbrace{\eta_i}_{=\sum l_{ik}\beta_k} - \underbrace{\left(\frac{\mathbf{u}_i}{\mathbf{W}_{ii}} + \hat{\eta}_i \right)}_{z_i} \right)^2 \right] \quad (59)$$

$$\vdots$$

$$= \sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left((z_i^k)^2 - 2z_i^k l_{ik} \beta_k + l_{ik}^2 \beta_k^2 \right) \right]. \quad (60)$$

Also note the following regarding the prior $g(\beta)$ which will be used later on:

- $\underline{\beta}_{k \times 1} \sim N(0, \sigma^2 I_{k \times k})$ (vector)
- $\beta \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$ (k-th β)
- $g(\underline{\beta}) = \prod_k g(\beta_k) = \prod_k \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2} \beta_k^2\right) \right]$ (where the quantity inside the brackets is the density of $g(\beta)$)
- Applying a log to above yields: $\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2} \sum_k \beta_k^2\right)$

We also note an expectation-variance equivalence as follows:

$$E_{q_l} [l_{ik}^2] = \text{Var}_{q_l} [l_{ik}] + (E_{q_l} [l_{ik}])^2 \quad (61)$$

such that

$$\overline{l_{ik}^2} = \text{Var}_{q_l} [l_{ik}] + (\overline{l_{ik}})^2 \quad (62)$$

because

$$E_{q_l} [l_{ik}] = \overline{l_{ik}} \text{ and } E_{q_l} [l_{ik}^2] = \overline{l_{ik}^2} \quad (63)$$

Recall,

$$\sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(\eta_i - \left(\frac{\mathbf{u}_i}{\mathbf{W}_{ii}} + \hat{\eta}_i \right) \right)^2 \right] \quad (64)$$

where $\eta_i = \sum_k l_{ik} \beta_k$ and $z_i = \frac{\mathbf{u}_i}{\mathbf{W}_{ii}} + \hat{\eta}_i$.

Then consider the following for the objective function for updates of q_β :

$$F(q_\beta, g_\beta) = E_{q_l, q_f, q_\beta} \left[\sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(z_i - \sum_k l_{ik} \beta_k \right)^2 \right] \right] \quad (65)$$

$$= E_{q_l, q_f, q_\beta} \left[\sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(z_i^2 - 2z_i \sum_k l_{ik} \beta_k + \left(\sum_k l_{ik} \beta_k \right)^2 \right) \right] \right] \quad (66)$$

$$\text{where } \left(\sum_k l_{ik} \beta_k \right)^2 = \sum_k l_{ik}^2 \beta_k^2 + 2 \sum_{k \neq k'} l_{ik} \beta_k l_{ik'} \beta_{k'} \quad (67)$$

$$= E_{q_l, q_f, q_\beta} \left[\sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(z_i^2 - 2z_i \sum_k l_{ik} \beta_k + \sum_k l_{ik}^2 \beta_k^2 + 2 \sum_{k \neq k'} l_{ik} \beta_k l_{ik'} \beta_{k'} \right) \right] \right] \quad (68)$$

Now, treating L as a random variable (apply expectation of q_l , retain q_β) we get

$$F(q_\beta, g_\beta) = E_{q_l, q_f, q_\beta} \left[\sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(z_i^2 - 2z_i \sum_k \bar{l}_{ik} \beta_k + \dots \right) \right] \right] \quad (69)$$

$$= E_{q_l, q_f, q_\beta} \left[\sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(-2z_i \sum_k \bar{l}_{ik} \beta_k + \sum_k \bar{l}_{ik}^2 \beta_k^2 + 2 \sum_{k \neq k'} \bar{l}_{ik} \bar{l}_{ik'} \beta_k \beta_{k'} \right) \right] \right] \quad (70)$$

$$+ E_{q_\beta} \left[\log \frac{g_\beta(\beta)}{q_\beta(\beta)} \right] \\ = E_{q_l, q_f, q_\beta} \left[\sum_i \left[-\frac{1}{2} \mathbf{W}_{ii} \left(-2z_i \sum_k \bar{l}_{ik} \beta_k + \sum_k \beta_k^2 (\bar{l}_{ik})^2 + 2 \sum_{k \neq k'} \bar{l}_{ik} \bar{l}_{ik'} \beta_k \beta_{k'} \right) \right] \right] \quad (71)$$

$$+ E_{q_l, q_f, q_\beta} \left[\sum_i -\frac{1}{2} \mathbf{W}_{ii} \left(\sum_k \text{Var}_{q_l}[l_{ik}] \beta_k^2 \right) \right] + E_{q_\beta} [\log g(\beta)] - E_{q_\beta} [\log q(\beta)] \\ = E_{q_l, q_f, q_\beta} \left[\sum_i -\frac{1}{2} \mathbf{W}_{ii} \left(z_i - \sum_k \bar{l}_{ik} \beta_k \right)^2 \right] \quad (72)$$

$$+ E_{q_l, q_f, q_\beta} \left[\sum_i -\frac{1}{2} \mathbf{W}_{ii} \left(\sum_k \text{Var}_{q_l}[l_{ik}] \beta_k^2 \right) \right] + E_{q_\beta} [\log g(\beta)] - E_{q_\beta} [\log q(\beta)]$$

This holds because the expansion of the square $(z_i - \sum_k \bar{l}_{ik} \beta_k)^2$ yields

$$\left(z_i - \sum_k \bar{l}_{ik} \beta_k \right)^2 = z_i^2 - 2z_i \sum_k \bar{l}_{ik} \beta_k + \sum_k (\bar{l}_{ik})^2 \beta_k^2 + 2 \sum_{k \neq k'} \bar{l}_{ik} \bar{l}_{ik'} \beta_k \beta_{k'} \quad (73)$$

which matches the previous expanded term.

Now consider the distribution of

$$g(\beta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{1}{2\sigma^2} \sum_k \beta_k^2 \right] \quad (74)$$

such that

$$E_{q_\beta} [\log g(\beta)] = E_{q_\beta} \left[\log \left[\frac{1}{\sqrt{1\pi\sigma^2}} \exp \left(-\frac{1}{2\sigma^2} \sum_k \beta_k^2 \right) \right] \right] \quad (75)$$

$$= E_{q_\beta} \left[\log \left(\frac{1}{\sqrt{1\pi\sigma^2}} \right) - \frac{1}{2\sigma^2} \sum_k \beta_k^2 \right] \quad (76)$$

$$= -\frac{1}{2\sigma^2} E_{q_\beta} \left[\sum_k \beta_k^2 \right] \quad (77)$$

Substituting this result into the full expression yields the following:

$$F(q_\beta, g_\beta) = E_{q_l, q_f, q_\beta} \left[\sum_i -\frac{1}{2} \mathbf{W}_{ii} \left(z_i - \sum_k \bar{l}_{ik} \beta_k \right)^2 \right] \\ + E_{q_l, q_f, q_\beta} \left[\sum_i -\frac{1}{2} \mathbf{W}_{ii} \left(\sum_k \text{Var}_{q_l}[l_{ik}] \beta_k^2 \right) \right] - E_{q_\beta} \left[\frac{1}{2\sigma^2} \sum_k \beta_k^2 \right] - E_{q_\beta} [\log q(\beta)] \quad (78)$$

Then consider just the following terms:

$$E_{q_l, q_f, q_\beta} \left[\sum_i -\frac{1}{2} \mathbf{W}_{ii} \left(\sum_k \text{Var}_{q_l}[l_{ik}] \beta_k^2 \right) \right] - E_{q_\beta} \left[\frac{1}{2\sigma^2} \sum_k \beta_k^2 \right] \quad (79)$$

$$= E_{q_l, q_f, q_\beta} \left[\sum_i -\frac{1}{2} \mathbf{W}_{ii} \left(\sum_k \text{Var}_{q_l}[l_{ik}] \beta_k^2 \right) - \frac{1}{2\sigma^2} \sum_k \beta_k^2 \right] \quad (80)$$

$$= E_{q_l, q_f, q_\beta} \left[\sum_k \left(\sum_i -\frac{1}{2} \mathbf{W}_{ii} \text{Var}_{q_l}[l_{ik}] \right) \beta_k^2 - \frac{1}{2\sigma^2} \sum_k \beta_k^2 \right] \quad (81)$$

$$= E_{q_l, q_f, q_\beta} \left[-\frac{1}{2} \sum_k \left(\sum_i \mathbf{W}_{ii} \text{Var}_{q_l}[l_{ik}] + \frac{1}{\sigma^2} \right) \beta_k^2 \right] \quad (82)$$

Therefore the full objective function to minimize for updates of q_β is

$$\begin{aligned} F(q_\beta, g_\beta) &= E_{q_l, q_f, q_\beta} \left[-\frac{1}{2} \sum_i \mathbf{W}_{ii} \left(z_i - \sum_k \bar{l}_{ik} \beta_k \right)^2 \right] \\ &\quad + E_{q_l, q_f, q_\beta} \left[-\frac{1}{2} \sum_k \left(\sum_i \mathbf{W}_{ii} \text{Var}_{q_l}[l_{ik}] + \frac{1}{\sigma^2} \right) \beta_k^2 \right] - E_{q_\beta} [\log q(\beta)] \end{aligned} \quad (83)$$

Then consider a weighted least squares regression problem where

$$z_i = \sum_k X_{ik} \beta_k + \epsilon_i \quad (84)$$

where

- $X_{ik} = \bar{l}_{ik}$ such that $z_i = \sum_k \bar{l}_{ik} \beta_k + \epsilon_i$
- $\epsilon_i \sim N\left(0, \frac{1}{\mathbf{W}_{ii}}\right)$
- $g(\beta_k) \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

It follows that

$$F(q_\beta, g_\beta) = E_{q_\beta} [\log P(\underline{Z}|\underline{\beta})] + E_{q_\beta} \left[\log \frac{g(\underline{\beta})}{q(\underline{\beta})} \right] \quad (85)$$

$$= E_{q_l, q_f, q_\beta} \left[-\frac{1}{2} \mathbf{W}_{ii} \left(z_i - \underbrace{\sum_k X_{ik} \beta_k}_{\text{likelihood}} \right)^2 - \underbrace{\frac{1}{2\sigma^2} \sum_k \beta_k^2}_{\text{prior}} - \underbrace{\log q_\beta(\beta)}_{\text{posterior}} \right] \quad (86)$$

Now formulate a weighted least squares regression problem and solve the updates to β with software.

simpler version is EBMF (assumes independence)

6A Update for q_β - EBMF focus (vs. WLS)

6A.1 Setting Up the Objective

Because β appears only in the survival term, we drop the genomics log-likelihood (which is constant with respect to β) and focus on the survival likelihood and the prior/posterior terms. The full objective function for q_β is:

$$F(q_\beta, g_\beta) = E_{q_l, q_\beta} [\log P(t, \delta | L, \beta)] + \sum_k E_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \quad (87)$$

Using the second-order Taylor approximation from the q_l update, we substitute the working response z_i and weights W_{ii} :

$$F(q_\beta, g_\beta) \approx E_{q_l, q_\beta} \left[-\frac{1}{2} \sum_i W_{ii} \left(z_i - \sum_k l_{ik} \beta_k \right)^2 \right] + \sum_k E_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \quad (88)$$

6A.2 Coordinate-Ascent Update for β_k

By applying the mean-field independence assumption $q(\beta) = \prod_k q(\beta_k)$, we can update a single coefficient β_k by fixing all $\beta_{k'}$ for $k' \neq k$. We define the survival partial working response, which represents the residual survival signal after removing the expected effects of all other factors:

$$z_i^{-k} = z_i - \sum_{k' \neq k} \overline{l_{ik'}} \beta_{k'} \quad (89)$$

Substitute this partial response decomposition, $z_i \approx z_i^{-k} + l_{ik} \beta_k$, into the objective function's squared term:

$$-\frac{1}{2} \sum_i W_{ii} (z_i^{-k} - l_{ik} \beta_k)^2 = -\frac{1}{2} \sum_i W_{ii} ((z_i^{-k})^2 - 2z_i^{-k} l_{ik} \beta_k + l_{ik}^2 \beta_k^2) \quad (90)$$

Now, we take the expectation with respect to the variational posteriors q_l and q_β . The term $(z_i^{-k})^2$ is constant with respect to β_k and drops into a constant C . Critically, we apply the expectation over L to the linear and squared terms, using $E_{q_l}[l_{ik}] = \overline{l_{ik}}$ and $E_{q_l}[l_{ik}^2] = \overline{l_{ik}^2}$:

$$\begin{aligned} F(q_{\beta_k}) = \sum_i & \left[W_{ii} z_i^{-k} \overline{l_{ik}} \beta_k - \frac{1}{2} W_{ii} \overline{l_{ik}^2} \beta_k^2 \right] \\ & - \frac{1}{2\sigma^2} \beta_k^2 + E_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] + C \end{aligned} \quad (91)$$

Note: The term $-\frac{1}{2\sigma^2} \beta_k^2$ originates from the prior $g(\beta_k) \sim \mathcal{N}(0, \sigma^2)$. In the software implementation, this fixed prior variance term is subsumed into the EBNM prior estimation, as the solver empirically fits g_{β_k} directly from the data.

6A.3 Formulating the EBNM Problem

We group the remaining terms to identify the coefficients for the standard quadratic form $-\frac{1}{2} A_k \beta_k^2 + B_k \beta_k$:

scalar!

$$A_k = \sum_i W_{ii} \overline{l_{ik}^2} + \frac{1}{\sigma^2} \quad (92)$$

$$B_k = \sum_i W_{ii} z_i^{-k} \overline{l_{ik}} \quad (93)$$

Using the full second moment $\overline{l_{ik}^2} = \text{Var}_q(l_{ik}) + \overline{l_{ik}}^2$ rather than just the squared mean $\overline{l_{ik}}^2$ in A_k encodes an error-in-variables correction. The uncertainty in L appropriately inflates the effective noise for β_k , preventing overfitting.

This yields the inputs for the 1D EBNM problem with a single pseudo-observation for factor k :

$$x_k = \frac{B_k}{A_k} = \frac{\sum_i W_{ii} z_i^{-k} \overline{l_{ik}}}{\sum_i W_{ii} \overline{l_{ik}^2} + 1/\sigma^2} \quad (94)$$

$$s_k = \frac{1}{\sqrt{A_k}} \quad (95)$$

For each factor k , we solve the EBNM problem:

$$(\hat{g}_{\beta_k}, \hat{q}_{\beta_k}) = \text{EBNM}(x_k, s_k) \quad (96)$$

7 Update for τ

We derive updates to optimize the objective function F over the precision parameters τ identically to the procedure defined in section A.1 of the EBMF paper (equations (A.5) – (A.11)). This update specifically focus on the components of F that depend on τ such that

$$F(\tau) = E_{q_l} E_{q_f} \sum_{ij} \frac{1}{2} \log(\tau_{ij}) - \frac{1}{2} \tau_{ij} \left(Y_{ij} - \sum_k l_{ik} f_{jk} \right)^2 + C \quad (97)$$

$$= \frac{1}{2} \sum_{ij} \left[\log(\tau_{ij}) - \tau_{ij} \overline{R^2}_{ij} \right] + C \quad (98)$$

where $\overline{R^2}_{ij}$ is defined by

$$\overline{R^2}_{ij} := E_{q_l, q_f} \left[\left(Y_{ij} - \sum_{k=1}^K l_{ik} f_{jk} \right)^2 \right] \quad (99)$$

$$= \left(Y_{ij} - \sum_k \bar{l}_{ik} \bar{f}_{jk} \right)^2 - \sum_k (\bar{l}_{ik})^2 (\bar{f}_{jk})^2 + \sum_k \bar{l}_{ik}^2 \bar{f}_{jk}^2 \quad (100)$$

If $\tau \in \mathcal{T}$ is constrained, we have

$$\hat{\tau} = \arg \max_{\tau \in \mathcal{T}} \sum_{ij} \left[\log(\tau_{ij}) - \tau_{ij} \overline{R^2}_{ij} \right] \quad (101)$$

Assuming constant precision $\tau_{ij} = \tau$ yields

$$\hat{\tau} = \frac{NP}{\sum_{ij} \overline{R^2}_{ij}} \quad (102)$$

and assuming column-specific precisions ($\tau_{ij} = \tau_j$) yields

$$\hat{\tau}_j = \frac{N}{\sum_i \overline{R^2}_{ij}} \quad (103)$$

In equations 102 and 103, the variables N and P refer to the following:

- N : The number of samples/rows in the data matrix Y
- P : The number of features/columns in the data matrix Y

Therefore, for equation 102, NP refers to the total number of entries in the $n \times p$ matrix. This equation sets the precision to the inverse of the average expected squared residual across all N rows and P columns. Furthermore, for equation 103, N , the number of rows, is used because the precision for a specific column is estimated based on the N observations within that column.